

CPCS203 MID23

MCQ:

1- What is the output of the following code? [0.5 Mark]

```
String s1 = "Ramadan Mubarak!";  
String s2 = "Eid Mubarak!";  
System.out.println(s1.compareTo(s2));
```

- A) positive number
- B) 0
- C) negative number
- D) false

2- What is the output of the following code? [0.5 Mark]

```
String s1 = "CPCS203 Midterm Exam" ;  
String s2 = new String ("CPCS203 Midterm Exam") ;  
System.out.print(s1.equals(s2) + " ");  
System.out.println(s1 == s2)
```

- A) false true
- B) true false
- C) false false
- D) true true

3- What is the output of the following code? [0.5 Mark]

```
public static void main (String args[]) {  
    String str = "Keep, Calm and Love? Java. Programming";  
    String [] arrOfStr = str.split("[, ?.@]+");  
    for (String a : arrOfStr)  
        System.out.print(a + " ");  
}
```

- A) Keep, Calm and@Love?Java.Programming
- B) KeepCalmAndLoveJavaProgramming
- C) Keep Calm and Love Java Programming
- D) Keep+Calm+and+Love+Java+Programming

4-What is the out put of the following code

```
System.out.print ("Chat G, P;T", replaceAll (" , ;", "#") + " ");
```

- A) Chat G P T
- B) Chat G#P#T
- C) Chat G,;#P,;#T
- D) Chat G,P;T

What is the output of the following code? [2 Marks]

```
public class Test 1 {  
    public static void main (String args []) {  
        StringBuilder sb1 = new StringBuilder ("It was in the month of") ;  
        StringBuilder sb2 = new StringBuilder(" that the Quran was revealed  
        ");  
        sb1.append (sb2) ;  
        System.out.println(sb1) ;  
        StringBuilder sb3 = new StringBuilder("nadamar") ;  
        sb3.reverse();  
        System.out.println (sb3);  
        sb3.setCharAt (0, 'R');  
        System.out.println (sb3) ;  
        StringBuilder sb5 = new StringBuilder(sb1.insert(22, " " + sb3));  
        System.out.println(sb5) ;  
        StringBuilder sb6 = new StringBuilder (" as guidance for mankind") ;  
        sb5.append (sb6) ;  
        System.out.println(sb5) ;  
    }  
}
```

Output:

It was in the month of that the Quran was revealed
ramadan
Ramadan
It was in the month of Ramadan that the Quran was revealed
It was in the month of Ramadan that the Quran was revealed as guidance for mankind

Questions regarding file/io

Complete the following code by creating a file, named TheoreticalExams.txt writing in it what appears below. Assume that the file does not exist.

TheoreticalExams.txt

Amal 19 28

```
Import java.util.Scanner;
import java.io.*; or
import java.io.File;
import java.io.FileNotFoundException;
import java.io.PrintWriter;
Public class writeOnFile {
    Public static void main(String [] args) throws Exception {

        File file = new File("TheoreticalExams.txt");
        PrintWriter pen = new PrintWriter(file);
        pen.println("Amal 19 28");
        pen.flush();
        pen.close();

    }
}
```

Assuming that the above file TheoreticalExams.txt exists, read from it the midterm score, and final exam score, then print them on the console.

```
3 import java.io.*;
4 import java.util.Scanner;
5
6 public class Mid23 {
7
8     public static void main(String[] args) throws FileNotFoundException{
9         File file = new File("TheoreticalExams.txt");
10        Scanner read = new Scanner(file);
11        double midScore = read.nextDouble();
12        double finalScore = read.nextDouble();
13        System.out.println("The mid score is " + midScore);
14        System.out.println("The final score is " + finalScore);
15    }
16 }
```

What is the output of the following statement?

```
File f = new File ("c:\\user\\KAU\\FCIT\\..\\CPCS203\\MidtermExam.txt")
System.out.println("File name: " + f.getCanonicalPath ())
System.out.println("is File: " + f.isFile ());
```

File name: c:\user\KAU\CPCS203\MidtermExam.txt
is File: true

What is the output of the following code?

```
public class PublishBooks {

    public static void main(String [] args) {
        Book book1 = new Book ();
        System.out.println(book1.author);
        System.out.println (book1.isPublished);
    }
}

class Book {
    public String bookTitle;
    public String author;
    public boolean isPublished;
}
```

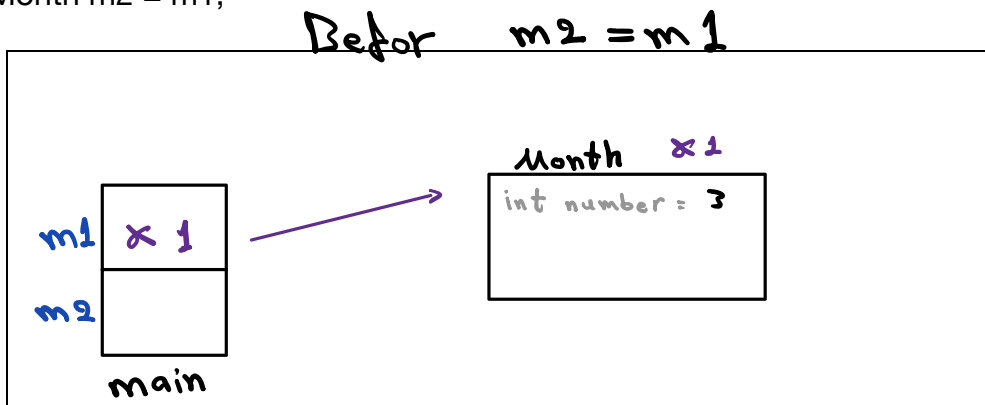
null
false

Given the code below, explain by drawing or by writing what is happening between objects

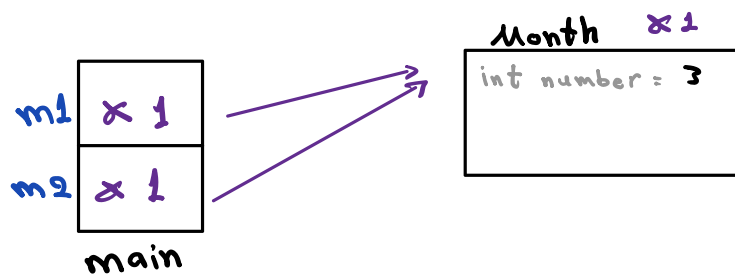
m1 and m2

Month m1 = new Month (3) ;

Month m2 = m1;



After m2 = m1



Q4 Draw the UML diagram of class Employee and indicate whether it is immutable or not and justify your answer.

```
class Employee{
private String name;
int id;
public String department;
public static int numberOfEmployees;

public Employee (String newName, int ID, String dept, int numOfEmp){
name = newName;
id = ID;
department = dept;
numberorEmployees = numOfEmp;
}

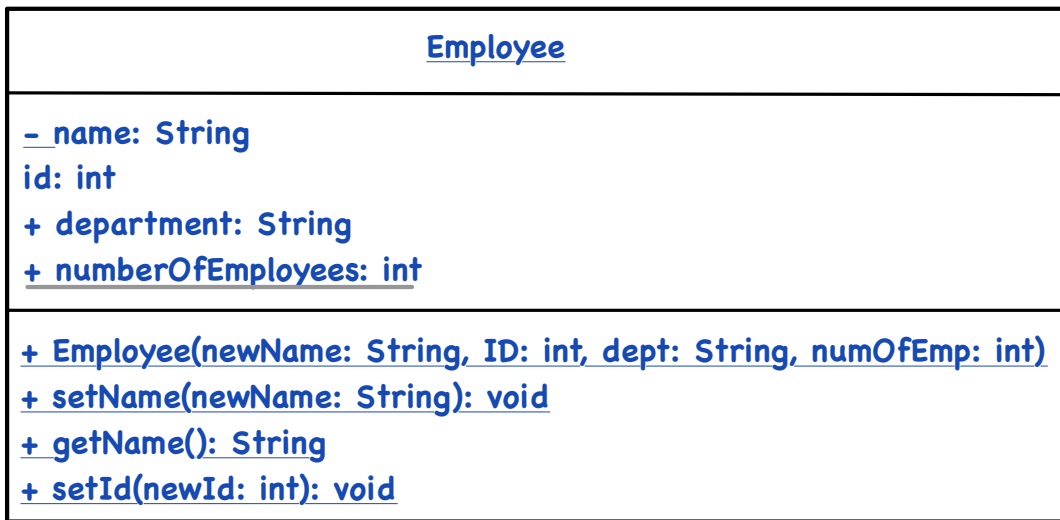
public void setName (String newName){
name = newName;
}

public String getName (){
return name;
}
public void setId (int newId) {
Id = newId,
}
```

Is class Employee immutable? Justify.

No because some of the data fields is public so we can change it after the initialise from the another class

DRAW THE UML HERE



Q5. Write the missing parts of the code according to the below requirements.

The Traffic Violation System helps the traffic department to register vehicles and violation details. After registration, the system issues a violation ticket in case of invalid routine checks, or underage driver. The next three steps explain how to develop the system.

1. Complete the Violation class below by adding the following:

- A constructor that initializes the violation code and amount,
- Getters for the data field variables.

Violation Class:

```
public class Violation {  
    protected String code;  
    protected double amount;
```

```
    public Violation (String code, double amount){  
        this.code = code;  
    } this.amount = amount;
```

```
    public String getCode(____) {  
        return code;  
    }
```

```
    public double getAmount(____){  
        return amount;  
    }  
}
```


2. Complete the Vehicle class below by adding the following:

- 4 private instance variables:
- plate_number (string variable e.g., ANG-3456)
- driver's national_ID (int value e.g., 2342114)
- Age (int value e.g., 24)
- violation information(vi (an object of class Violation).
- A constructor that initializes all variables.
- A method called CalcAmount that calculates and returns the total violation amount. To calculate the total violation amount you should add additional penalty of 1000 SAR to the violation amount if the driver's age is less than 21 years.

Vehicle Class:

```
1. public class Vehicle {
2.     private String plate_number;
3.     private int nationalID;
4.     private int age;
5.     private Violation violation;
6.     public Vehicle (String p, int ID, int a, string code, double amount){
7.         plat_number = p;
8.         nationalID = ID;
9.         age = a;
10.        violation = new Violation(code, amount);
11.    }
12.    public double CalcAmount () {
13.        double amountPayment= violation.getAmount;
14.        if(age < 21)
15.            amountPayment += 1000;
16.        return amountPayment;
17.    }
18.    public String getPlate () {
19.        return plate number;
20.    }
21. }
```

3. Complete the main method below that has & vehicles.

All vehicles details will be registered into the system, You need to display the vehicle plate number and total penalty amount that should be paid by each vehicle.

```
1. public class Test{
2. public static void main (String args []) {
3.
4. System.out.println ("Please enter the number of cars: ");
5. int n= input.nextInt();
6. // Create an array of Vehicles (n Vehicles) with the reference cars
7. Vehicle[] vehicles = new Vehicle[n];
8. // after reading n Cars details from the user, you should create the objects in the
   array
9. for ( int i= 0;i< n; i++){
10. System.out.println("Please enter the car's plate number: ");
11. String P = input.nextln();
12. System.out.println ("Please enter the driver's national
   ID number: ");
13. int ID = input.nextInt();
14. System.out.println("Please enter the driver's age ");
15. int a = input.nextInt() ;
16. System.out.println("Please enter the violation code: ");
17. String code = input.nextln() ;
18. System.out.println ("Please enter the violation amount: ");
19. double amount = input.nextDouble;
20.
21. vehicles[i] = new Vehicle(P, ID, a, code, amount);
22. }
23. // calculate and display for each vehicle the plate number and violation amount that
   should be pard
24. for ( int i= 0;i< n; i++) {
25. sout("the vehicle plat number is " + vehicles[i].getPlat() +
   -- " and valuation amount is " + vehicles[i].CalculatAmount());
26. }
27. }
```

Q6(Inheritance)

```
class Bicycle {
    private int gear;
    private int speed;

    public Bicycle (int bgear, int bspeed) {
        gear = bgear;
        speed = bspeed;
    }

    public String toString() {

        return ("No of gears are " + gear + " speed is " + speed);
    }
}

class MountainBike extends Bicycle {
    private int sHeight;

    public MountainBike (int gear, int speed, int startHeight)
    // invoking the Bicycle constructor
    super(gear, speed);
    // initializing the sHeight
    sHeight = startHeight;
}

@Override public String toString () {
    return ( super.toString() + " seat height is " + sHeight);
}
}
```